Inherited Risk for Prostate Cancer Spurs Clinical Trial Participation

Joe

Thanks to an NCI research study, Joe's prostate cancer was found at an early stage.

Credit: Photo courtesy of Joe

A preponderance of cancers among family members fueled Joe's concern that he may have a genetic predisposition for cancer. He decided to have genetic testing in 2003, and the testing showed that he carried a BRCA2 mutation.

While BRCA mutations are best known for their association with an increased risk of breast and ovarian cancers in women, alterations in BRCA genes can also increase a man's risk for prostate cancer. Men who have inherited alterations in BRCA and other genes are much more likely to be diagnosed at a younger age and have prostate cancer that is aggressive. This knowledge spurred Joe to schedule annual appointments with doctors, consult a geneticist, and join social networks.

A new tool for diagnosing prostate cancer

While participating in an online group for men with BRCA mutations in early 2019, Joe heard about an NCI study to develop better ways of detecting prostate cancer early in highrisk populations. Men with BRCA and other high-risk genetic alterations who enroll are screened for prostate cancer using multiparametric magnetic resonance imaging (mpMRI).

It is current practice to use a PSA test to aid prostate cancer screening. Yet, the correlation between PSA level and a prostate cancer diagnosis in men with BRCA2 or other mutations is unknown; there is no standard way to screen men with high-risk genetics for prostate cancer.

While MRI scans produce detailed images of organs and tissues in the body, mpMRI technology provides more precise scans that can help doctors find and biopsy prostate tumors far better than a standard MRI. The goal of the NCI study is to develop better ways of detecting prostate cancer early in high-risk populations.

The researchers also plan to follow enrollees over the years to learn more about the natural history and progression of prostate cancer.

Joining a prevention study enabled early detection

Joe enrolled in the study in spring 2019 and went to the National Institutes of Health (NIH) Clinical Center to meet the NCI clinical care team, undergo laboratory tests, and have an mpMRI scan. On the way to the airport to fly home to California, he received a call from the NCI clinical care team asking him to come back to the NIH Clinical Center. They saw something on his mpMRI scan.

Joe returned to NIH, learned he had a tumor, and agreed to have a biopsy. "I was blown away by the accuracy of the biopsy technique that NCI used. Cancer can be anywhere in the prostate and can be missed by a typical biopsy, which involves sampling multiple random areas. But in my case, we had the mpMRI scan. We knew where my small tumor was located."

Using the highly detailed scan, Joe’s doctors were able to pinpoint the exact location of the tumor and sample tissue from that area. Examination of the specimen revealed cancer.

Several months later, Joe elected to have his prostate gland surgically removed in California. The pathology revealed an aggressive prostate cancer confined to the gland. While Joe no longer comes to the NIH Clinical Center for mpMRI screening, he continues to follow up with his NCI care team by phone.

In 2021, Joe was back to training for a 500-mile, week-long cycling event to raise funds for people with arthritis. This was an activity he put on hold for 2 years due to his prostate cancer experience and the COVID-19 pandemic. And NCI continues to enroll men with BRCA2 mutations in the mpMRI study.

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